

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

3rd Semester of B. Pharm. Examination

University Theory Examination November 2017

PH231/PH219 Pharmaceutical Engineering-III

Date: 29/11/2017, Wednesday

Time: 10 a.m. to 01 p.m.

Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat labeled sketches wherever necessary.

SECTION – I

- Q 1** Attempt all questions. Each question is of one mark. 20
1. Fluid energy mill is work on the principle of _____.
A. Impact
B. Compression
C. Cutting
D. Attrition
 2. Based on impact _____ mill works.
A. Hammer mill
B. Fluid energy mill
C. Roller mill
D. Cutter mill
 3. _____ method is used for determination of particle size of suspension.
A. Sieving
B. Andersen pipette
C. Optical microscopy
D. Scanning electron microscope
 4. According to I.P specification moderately fine powder find between sieve No. _____.
A. 10/44
B. 22/60
C. 44/85
D. 105/120

5. Convective mixing is refer to
 - A. Macromixing
 - B. Micromixing
 - C. Shear mixing
 - D. Diffusive mixing
6. Which of blender is considered as tumble blenders?
 - A. Twin shell blender
 - B. Double cone blender
 - C. Cubical Shaped blender
 - D. All of the above
7. Colloid mill is used for the preparation of
 - A. Syrup
 - B. Elixir
 - C. Mouthwash
 - D. Emulsion
8. Beater is an part of which mixing equipment
 - A. Sigma Blade
 - B. Double cone
 - C. Ribbon
 - D. Planetary
9. _____ is designed to streamline and improve the accuracy and efficiency of the medication use process.
 - A. Machinery
 - B. Automation
 - C. C.P.U.
 - D. Controller
10. Computer process control does not measure _____
 - A. Operator error
 - B. Temperature
 - C. Pressure
 - D. Flow rate
11. Solvent change and salting out method are used for preparation of
 - A. Plate crystal
 - B. Needle crystal
 - C. Spherical crystal
 - D. Hexagonal crystal
12. Oslo Crystallizer is also known as _____ crystallizer
 - A. Vacuum
 - B. Swenson Walker
 - C. Krystal Cooling
 - D. Agitated tank

13. _____ type of crystals is held together by Vander Waals forces.
- A. Ionic
 - B. Molecular
 - C. Covalent
 - D. Metallic
14. In cyclone separator, the separation process depends on _____
- A. Size and density
 - B. Porosity
 - C. Melting point
 - D. Crystal type
15. The ratio of initial particle size to final particle size is defined as _____
- A. Reduction ratio
 - B. Kick's ratio
 - C. Rittingers ratio
 - D. Bond's ratio
16. _____ suggests that energy required in a size reduction process is proportional to the new surface area produced.
- A. Rittinger's theory
 - B. Kick's ratio
 - C. Bond's ratio
 - D. Reduction ratio
17. Which one of the following is the majorly find route of exposure to hazardous drugs in pharmaceutical industry?
- A. Inhalation of an aerosolized drug
 - B. Dermal absorption
 - C. Ingestion
 - D. Injection
18. A process of crystallization taking place from vapor of a substance is known as _____
- A. Sublimation
 - B. Distillation
 - C. Precipitation
 - D. Filtration
19. Elutriation method is mainly used for particles smaller than _____ nm.
- A. 50
 - B. 100
 - C. 1000
 - D. 10
20. Elutriation is a process of _____
- A. Size separation using a moving fluids
 - B. Size separation using a stationary fluids
 - C. Size reduction using mechanical forces
 - D. Size reduction using electrical repelling forces

SECTION – II

Q 2 Attempt any TWO of the following.

- | | | |
|---|--|----|
| A | State and explain Mier's supersaturation theory for crystallization. | 05 |
| B | Classify the crystallizer. Differentiate small scale crystallizer with large scale crystallizer. | 05 |
| C | Give Swenson walker crystallizer working principle and application with labelled diagram. | 05 |

Q 3 Attempt any TWO of the following.

- | | | |
|---|---|----|
| A | How does size reduction equipment selection will affect milling property?
Explain in brief about the difference between wet and dry milling. | 05 |
| B | Give construction, working and uses of ball mill with labelled diagram. | 05 |
| C | Give various factors influencing size reduction with suitable example. | 05 |

SECTION – III

Q 4 Attempt any FOUR of the following

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|---|--|----|
| A | Describe construction and working of cyclone separator with labelled diagram. | 05 |
| B | Describe standards of sieve. How powders are classified according to size as per Indian Pharmacopeias? | 05 |
| C | Describe principle, working and uses of fluid energy mill with labelled diagram. | 05 |
| D | Write a note on fire hazards. | 05 |
| E | What are safety precautions to be require in pharmacy laboratory? Describe. | 05 |
| F | Give the principle and importance of size separation in pharmacy. | 05 |

Q 5 Attempt any FOUR of the following.

- | | | |
|-------|---|----|
| A | Comment on: Vortex formation is affects the liquid mixing. Give the diagrammatical representation of suppression of vortex. | 05 |
| B | Enlist mixers for semisolids mixing. Give principle, labeled diagram and advantages of Colloid mill. | 05 |
| C (a) | Classify the solid mixer as per mechanism of solid-solid mixing. | 03 |
| (b) | Briefly explain V -shaped mixer working principle with diagram. | 02 |
| D | Classify liquid mixers. Give principle, working, labeled diagram and advantages of propeller mixers. | 05 |
| E | Classify industrial hazard. Briefly explain pharmaceutical hazards. | 05 |
| F | Describe in brief elements of automatic process control systems. | 05 |